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Case Study



Platform Partner unstop

Case Study 3: Bio-Based Materials and Circular Solutions for a Net-Zero Future

I. Background

For more than a century, the chemical industry has relied heavily on fossil fuel-derived resources to manufacture paints, coatings, polymers, adhesives, and specialty chemicals. While these materials have driven significant technological advancements, they have also contributed to greenhouse gas emissions, resource depletion and environmental challenges. Recent global events, including pandemics, geopolitical conflicts, trade restrictions and supply chain disruptions, have further highlighted the risks associated with dependence on imported petroleum-based raw materials, resulting in shortages, price volatility and supply insecurity. As industries move toward a net-zero future, there is an increasing need to replace fossil derived carbon with renewable carbon while promoting locally sourced and sustainable feedstocks. Renewable resources such as vegetable oils, agricultural residues, lignin, cellulose, algae, microorganisms, enzymes and forestry biomass offer significant potential to develop high-performance, low-carbon materials that enhance circularity and supply-chain resilience.

Asian Paints Limited has already advanced this journey through innovations such as Nilaya Naturals, renewable-content products, recycled-content solutions, waterborne technologies and low-VOC formulations. The next challenge is to identify newer biobased or bioderived materials, develop scalable, commercially viable and high-performance renewable materials that can effectively replace conventional fossil-derived alternatives and accelerate the transition to a sustainable, net-zero economy.

II. The Opportunity

The transition to a net-zero future offers an unprecedented opportunity to redesign the materials ecosystem using renewable and circular resources. Advances in green chemistry, biotechnology, synthetic biology, and materials engineering are enabling the development of bio-based polymers, additives, pigments, and specialty materials from renewable feedstocks such as biomass, agricultural residues, algae, and carbon dioxide.

Imagine a future where:

- Wall paints contain polymers derived entirely from renewable carbon.
- Bio-based resins outperform conventional petrochemical materials.
- Buildings act as long-term carbon storage systems.
- Products are designed for circularity, recyclability, renewability, and low-carbon impact.
- Locally sourced renewable raw materials reduce dependence on imports and improve supply resilience.
- Waste streams are transformed into high-value materials, creating both environmental and economic benefits.

The future belongs to innovations that successfully combine performance, affordability, scalability and sustainability to accelerate the transition toward a circular and net-zero economy.

III. Key Challenges to Address

Despite significant progress, several barriers continue to hinder large-scale adoption of renewable materials:

Current bio-based solutions face several barriers:

- Inconsistent feedstock quality
- Limited scalability
- Higher cost
- Performance limitations
- Durability concerns
- Limited industrial infrastructure
- Supply chain constraints
- Competition with food resources
- Complex processing technologies

Innovators must address these barriers while delivering solutions that are technically feasible, economically attractive, environmentally beneficial, and scalable.

IV. The scope of the Case Study

Participants are invited to develop and propose an innovative material, technology, product or process that leverages renewable carbon resources to create sustainable and scalable solutions for the coatings, materials and specialty chemicals industry. The solution may address:

- Renewable monomers and polymers
- Renewable Functional Additives
- Novel Bio-Based Pigments and Fillers
- Advanced Technologies
- Coatings and Specialty Materials
- This case study challenges participants to envision and develop the next generation of renewable and circular materials that can accelerate the transition toward a resilient, low-carbon, and net-zero future.
- Do not restrict your ideas to solutions from coatings industry only. Innovative and original ideas will be scored higher and will have a greater chance to be selected for the next round. Simply reproducing an AI generated idea without original thinking will greatly reduce the chance of qualification. Wherever any content is taken or generated via AI, it should be clearly mentioned.
- While a proof of concept or prototype is preferred and will be given extra points, extensive reading and coming up with an innovative idea or solution from a different industry or different areas of science will also enable the selection to the next round.

V. Questionnaire

1. Is your idea or solution in the area of raw material, monomer, polymer, additive or coating? What is the end application?
2. How will the material be sourced / synthesized, processed, or manufactured?
3. Is it equivalent to conventional fossil-derived materials? What are the expected performance benefits? Will it be a replacement for an existing material or a new functionality.
4. Give a perspective on cost (although cost will not be a barrier for the idea selection).
5. What technical, economic, regulatory and supply chain challenges must be addressed to enable large-scale implementation?
6. Give the life cycle analysis (cradle to grave) for the solution you have provided.

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